

A Generalized Framework for Narrative Shift Detection in News Media

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Abstract

News media narratives evolve continuously over time, often changing in framing, tone, and emphasis even when the underlying topic remains the same. Detecting such narrative shifts is critical for understanding public discourse, media bias, and policy influence. This project proposes a generalized, unsupervised framework for narrative shift detection that operates at the sentence level and leverages temporal contrastive learning to model narrative evolution. The approach avoids reliance on labeled data, fixed topic definitions, or prompt-based large language models, enabling scalability and interpretability across diverse news domains.

1 Introduction and Motivation

News media narratives evolve continuously over time. Even when articles discuss the same topic, the framing, tone, emphasis, and intent often change in response to political, regulatory, technological, or societal developments. These changes, referred to as *narrative shifts*, play a crucial role in shaping public opinion and influencing policy discourse.

Existing approaches to analyzing narrative evolution face significant limitations. Topic models often conflate topic presence with narrative framing, while supervised change detection methods require labeled data that is scarce, subjective, and expensive to obtain. Recently, large language model (LLM)-based approaches have been explored, but they suffer from scalability, reproducibility, and interpretability concerns.

This project aims to address these challenges by proposing a generalized, scalable, and interpretable framework for detecting narrative shifts in news media without relying on labeled data, fixed topic definitions, or prompt-based LLM inference.

2 Background and Related Work

Early work on temporal text analysis focused on topic modeling approaches such as Latent Dirichlet Allocation (LDA) and Dynamic Topic Models (DTM). While effective for discovering latent topics, these methods are sensitive to vocabulary changes and fail to capture shifts in narrative framing within the same topic.

Distance-based methods using cosine similarity or KL-divergence measure representational changes over time but lack temporal learning mechanisms and require manually chosen thresholds. Clustering-based approaches similarly fail to model gradual narrative evolution and often produce unstable results.

Supervised narrative change detection methods require annotated datasets, which are difficult to construct due to the subjective nature of narratives. Generative and LLM-based methods introduce additional concerns related to hallucination, evaluation reliability, and reproducibility.

Temporal Contrastive Learning (TCL) has recently emerged as a powerful paradigm for learning representations from unlabeled sequential data. Prior works such as TS-TCC and ConTempo demonstrate that temporal contrastive objectives can effectively model temporal structure without supervision. These insights motivate our choice of TCL for narrative shift detection.

3 Problem Definition and Objectives

The objective of this project is to detect **narrative shifts**, defined as changes in how a topic is framed or discussed over time, rather than changes in topic itself.

The proposed framework aims to:

- Detect narrative shifts without labeled data.
- Remain robust to changing entities and vocabulary.

- Operate across multiple topics and domains.
- Provide interpretable outputs explaining detected shifts.

4 Dataset Description

The framework operates on a corpus of news articles represented as:

- Publication date
- Full article text

Articles may originate from multiple news sources and span long time periods. No manual annotations or labels are required. Narrative evolution is learned purely from temporal structure in the data.

5 Proposed Methodology

This section presents a detailed, end-to-end methodology for detecting narrative shifts in news media. The proposed framework models how narratives evolve over time without relying on labeled data, fixed topic assignments, or entity-specific supervision. The pipeline is designed to be scalable, interpretable, and robust to changing vocabulary and entities.

5.1 Overview of the Pipeline

Given a large corpus of timestamped news articles, the system:

- decomposes articles into sentence-level units,
- embeds sentences with contextual awareness,
- softly associates sentences with semantic topics,
- aggregates narrative information temporally,
- learns temporal representations using contrastive learning,
- and detects narrative shifts as representation divergence over time.

Each stage is described in detail below.

5.2 Stage 1: Sentence Segmentation

Input: Raw news articles with publication dates.

Each article is segmented into individual sentences while preserving their original order. Narrative framing often changes within an article; therefore, sentence-level analysis provides finer granularity than document-level representations.

Standard sentence tokenization tools such as spaCy or NLTK are used. Each sentence is stored along with its article identifier and publication date.

Output: A temporally ordered sequence of sentences with metadata.

5.3 Stage 2: Context-Aware Sentence Embedding

Isolated sentences may be semantically ambiguous. To preserve contextual meaning, each sentence is embedded together with its neighboring sentences.

For a sentence s_i , the contextual input is defined as:

$$(s_{i-1}, s_i, s_{i+1})$$

Boundary tokens are used for the first and last sentences in an article. The contextual sentence triples are encoded using Sentence-BERT (SBERT), producing dense semantic representations:

$$\mathbf{s}_i \in \mathbb{R}^{768}$$

This approach preserves article-level coherence while enabling sentence-level narrative analysis.

Output: Contextual sentence embeddings for all sentences.

5.4 Stage 3: Semantic Topic Embedding Construction

Rather than treating topics as discrete labels, each topic is represented as a semantic region in the same embedding space as sentences.

For each topic k , a short natural language description is constructed that captures its semantic scope. This description is encoded using the same SBERT model:

$$\mathbf{T}_k \in \mathbb{R}^{768}$$

Encoding both sentences and topics using the same model enables direct semantic comparison without supervised classification.

Output: A set of topic embeddings $\{\mathbf{T}_1, \mathbf{T}_2, \dots, \mathbf{T}_K\}$.

5.5 Stage 4: Sentence-Level Topic Weighting

Each sentence embedding is softly associated with every topic using cosine similarity:

$$w_i^{(k)} = \cos(\mathbf{s}_i, \mathbf{T}_k)$$

Soft topic assignment preserves ambiguity and allows sentences to contribute to multiple overlapping narratives. This is essential for capturing gradual narrative transitions rather than abrupt topic switches.

Output: Topic relevance weights for each sentence.

5.6 Stage 5: Topic Filtering and Temporal Ordering

For a given topic k , sentences with relevance weight exceeding a predefined threshold τ are retained:

$$w_i^{(k)} \geq \tau$$

Retained sentences are sorted in non-decreasing order of publication date. Sentences may originate from different articles but are always temporally ordered to preserve narrative continuity.

Output: Topic-specific, temporally ordered sentence streams.

5.7 Stage 6: Same-Date Temporal Aggregation

Multiple articles published on the same date typically represent concurrent viewpoints rather than temporal evolution. To avoid spurious shifts, all topic-relevant sentences published on the same date are aggregated into a single narrative state.

For topic k on date d , aggregation is performed as:

$$\mathbf{Z}_{k,d} = \frac{1}{N_d} \sum_{i=1}^{N_d} \mathbf{s}_i$$

where N_d is the number of retained sentences on date d .

Output: One narrative representation per topic per date.

5.8 Stage 7: Topic-Wise Adaptive Temporal Windowing

Narratives evolve at different temporal rates across topics. Therefore, topic-specific window sizes W_k and strides S_k are defined.

Given daily narrative representations $\{\mathbf{Z}_{k,1}, \dots, \mathbf{Z}_{k,T}\}$, sliding windows are constructed as:

$$\mathcal{W}_t = [\mathbf{Z}_{k,t}, \mathbf{Z}_{k,t+1}, \dots, \mathbf{Z}_{k,t+W_k-1}]$$

Windows are temporally contiguous and may overlap depending on the stride S_k .

Output: A sequence of topic-specific temporal windows.

5.9 Stage 8: Temporal Contrastive Learning

The objective of Temporal Contrastive Learning (TCL) is to learn representations that distinguish narrative stability from narrative change without labeled annotations.

Adjacent windows form positive pairs:

$$(\mathcal{W}_t, \mathcal{W}_{t+1})$$

Temporally distant windows form negative pairs:

$$(\mathcal{W}_t, \mathcal{W}_{t+k}), \quad k \gg 1$$

A contrastive loss minimizes the distance between positive pairs while maximizing separation between negative pairs. This enables the model to learn narrative evolution directly from temporal structure.

Output: Temporally-aware window embeddings $\mathbf{Z}_t$.

5.10 Stage 9: Narrative Shift Scoring

Narrative shift magnitude is computed as the distance between consecutive window representations:

$$\Delta_t = \|\mathbf{Z}_t - \mathbf{Z}_{t-1}\|$$

Large values of Δ_t indicate significant narrative transitions, while small values correspond to narrative stability.

Output: A time series of narrative shift scores.

5.11 Stage 10: Interpretability and Explanation

To ensure interpretability, the framework identifies sentences that contribute most to detected shifts. This is achieved by analyzing changes in sentence-topic relevance or by sentence-level ablation.

The system returns top contributing sentences along with dates and article identifiers, enabling transparent inspection of narrative transitions.

5.12 Summary

The proposed methodology preserves semantic meaning at the sentence level, models narrative evolution through temporal contrastive learning, and detects narrative shifts without supervision. The design is scalable, robust to entity changes, and provides interpretable outputs suitable for real-world analysis.

6 Evaluation Strategy

Since labeled narrative shift datasets are unavailable, evaluation will be performed using:

- Qualitative case studies on well-known events.
- Temporal coherence analysis.
- Entity-centric narrative analysis.
- Sensitivity analysis across topics and window sizes.

These evaluations assess whether detected shifts align with real-world narrative transitions.

7 Tentative Timeline

Weeks 1–2	Dataset collection and preprocessing
Weeks 3–4	Sentence embedding and topic modeling
Weeks 5–6	Temporal aggregation and windowing
Weeks 7–8	Temporal contrastive learning implementation
Weeks 9–10	Evaluation and analysis
Weeks 11–12	Final report and presentation

8 Expected Contributions

This project proposes a novel, scalable, and domain-agnostic framework for narrative shift detection. By decoupling narrative modeling from fixed topics, labeled data, and LLM prompting, the approach offers a robust solution for analyzing narrative evolution in large-scale news corpora.

9 References Research Paper

- TS-TCC: Time-Series Representation Learning via Temporal and Contextual Contrast-ing. <https://arxiv.org/abs/2106.14112>

10 References Datasets

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